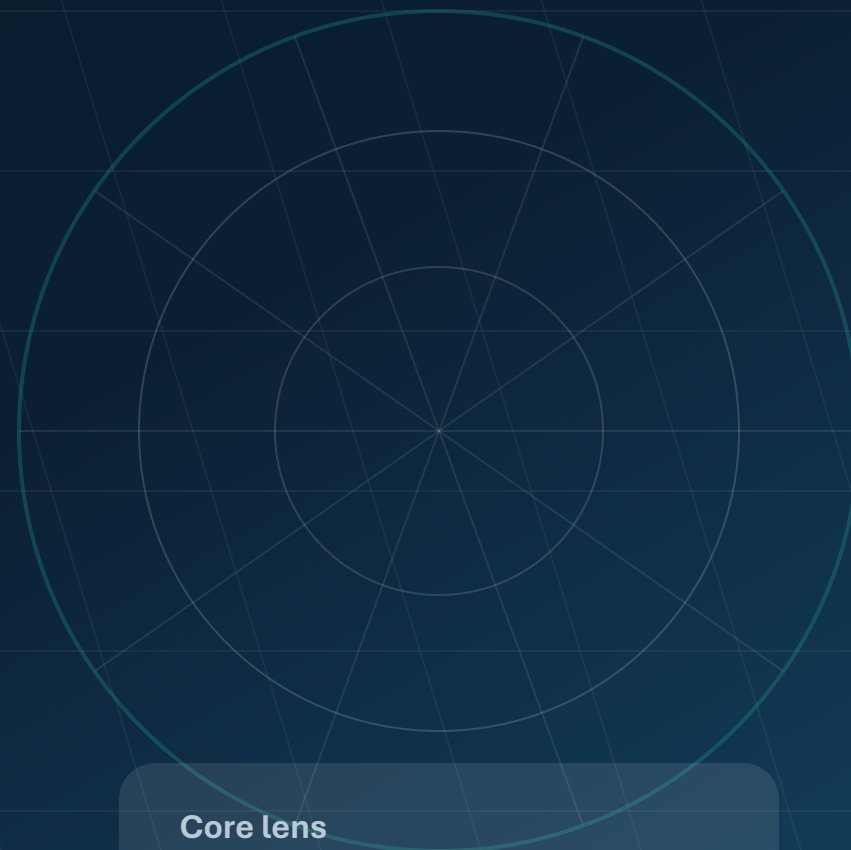


Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750–2024

Evidence briefing for policy analysts and climate negotiators

Data sources: NASA; Our World in Data; Global Carbon Budget 2025



Core lens

Totals × geography × per capita

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

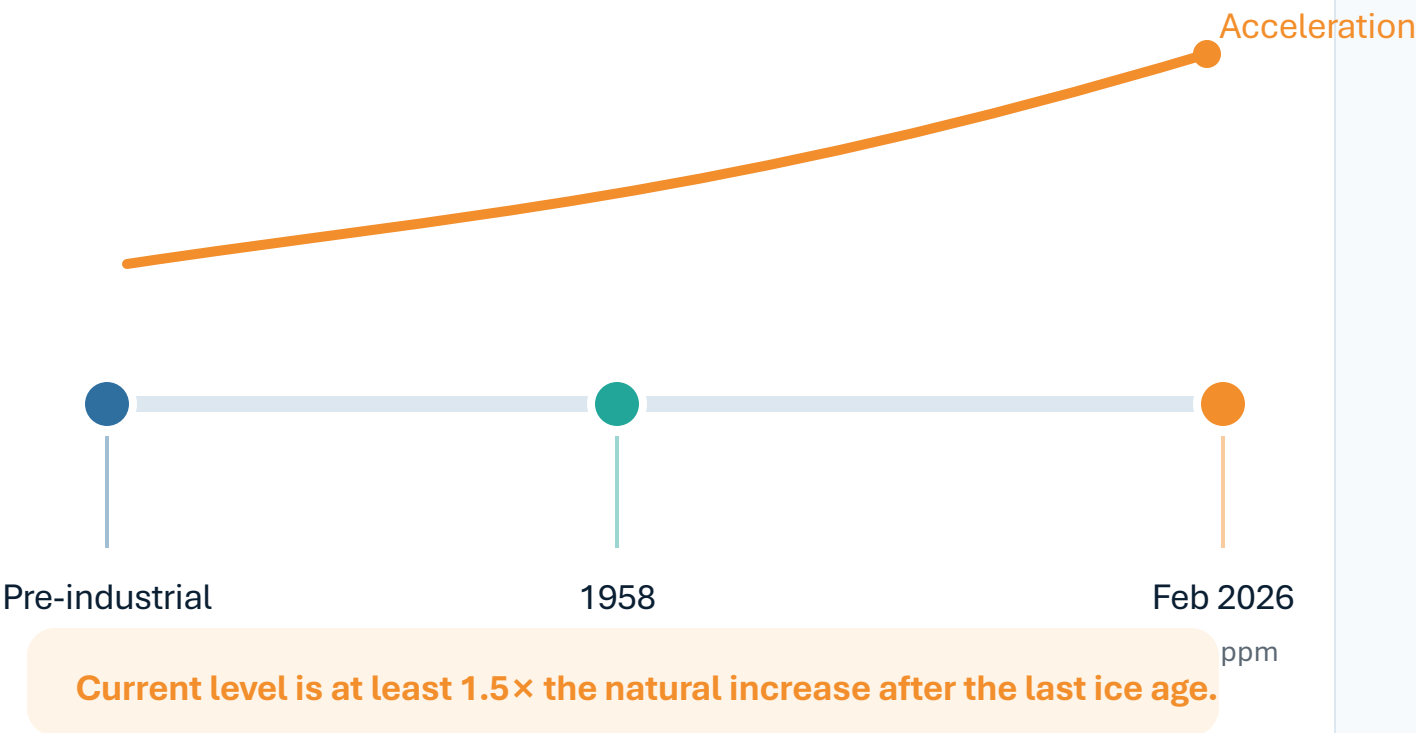
429 ppm

NOAA Mauna Loa Observatory, Feb 2026

>50% above ~280 ppm baseline

Atmospheric CO₂ is now more than 50% higher than the pre-industrial baseline. Ground-based and satellite measurements confirm sharp acceleration linked primarily to burning coal, oil, and natural gas.

A two-century jump after millennia-scale natural change



Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

Global annual fossil fuel & industry CO₂

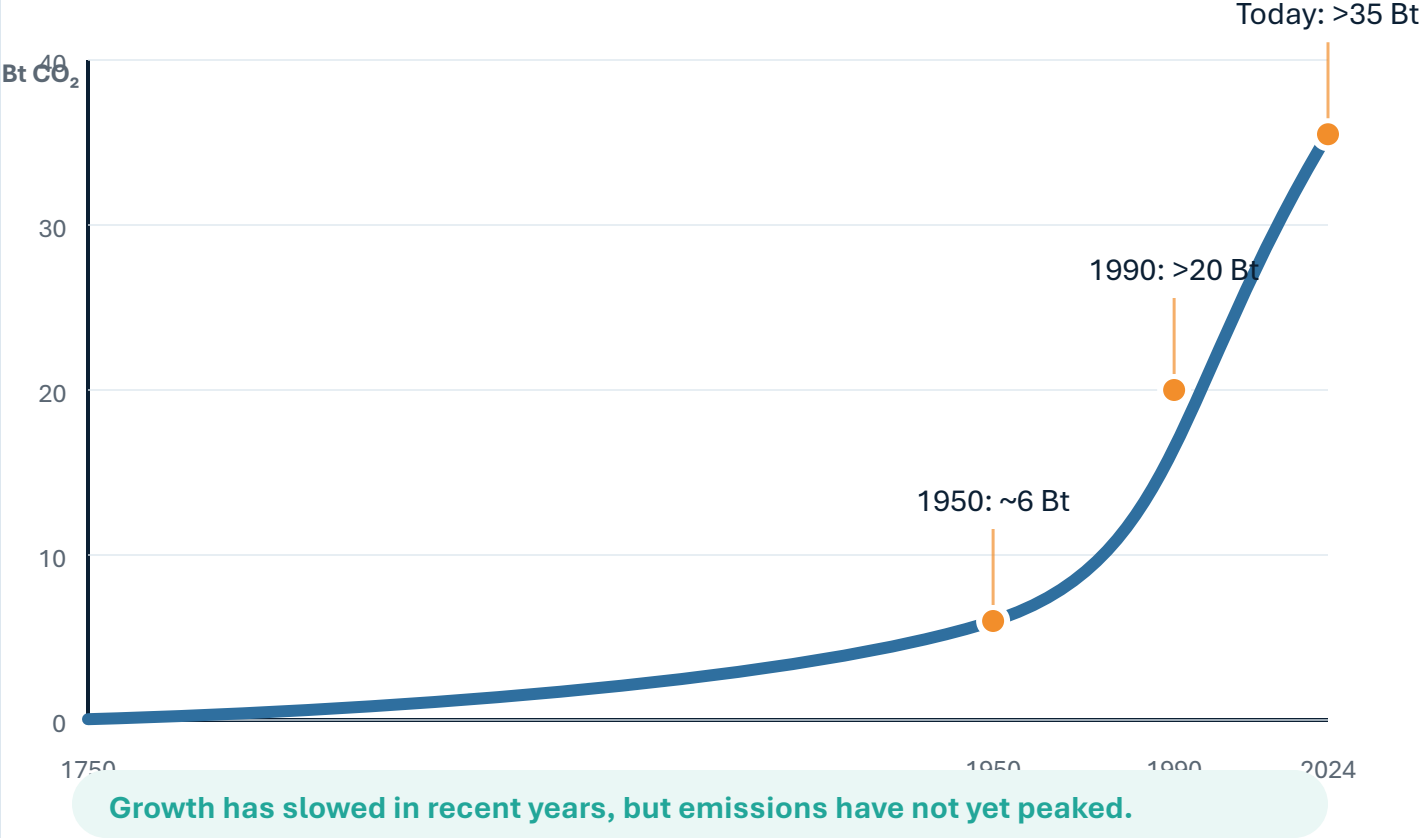
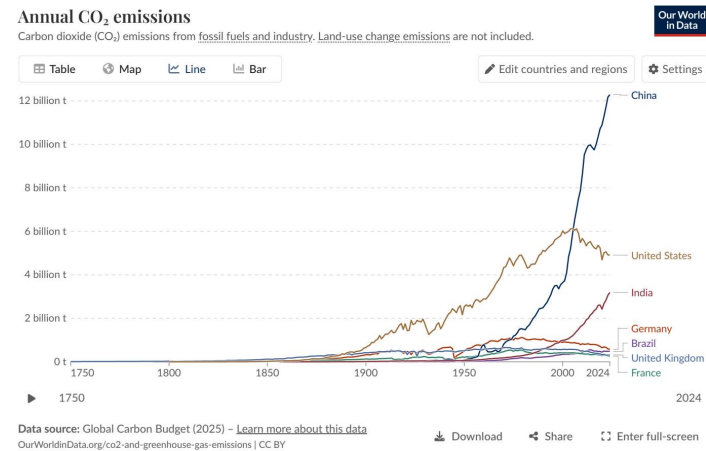


Figure reference

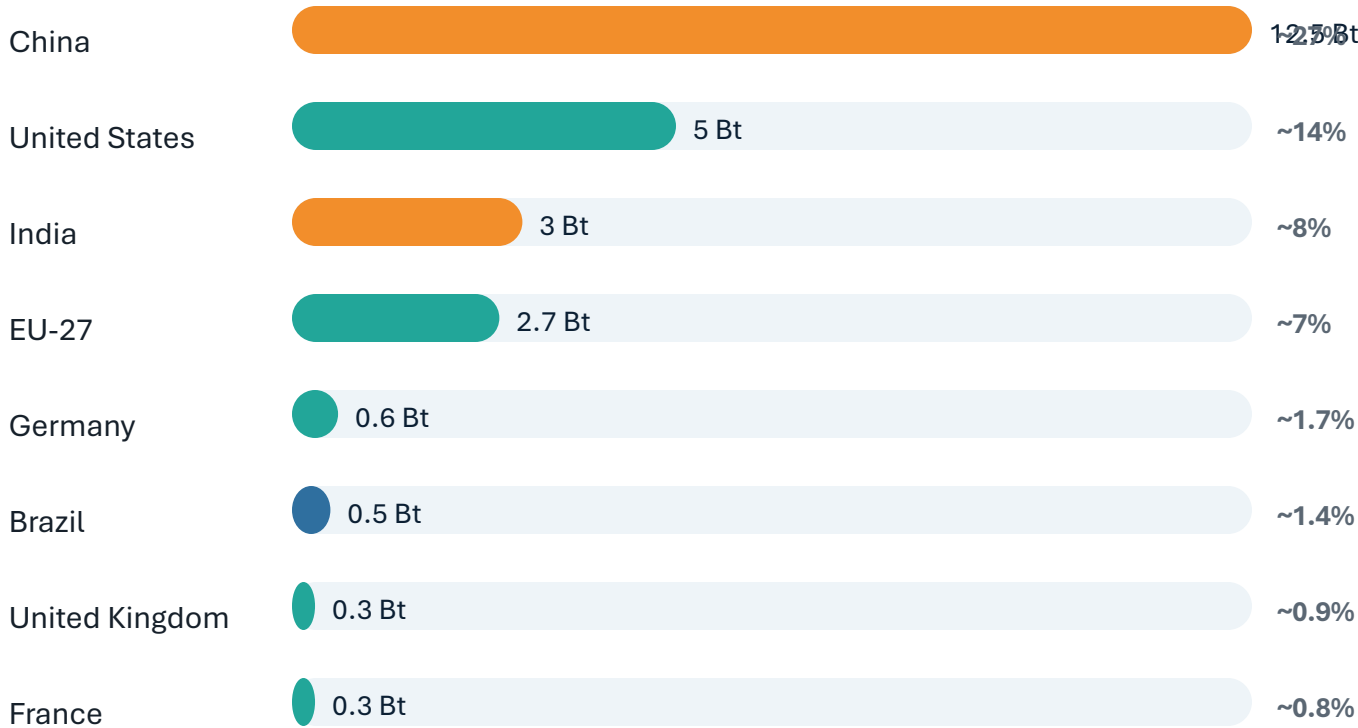


Annual CO₂ emissions by country contextualize the global growth path and the later national comparison.

Referenced: figure_3.jpg

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

Annual CO₂ emissions by country / region



Interpretation

China

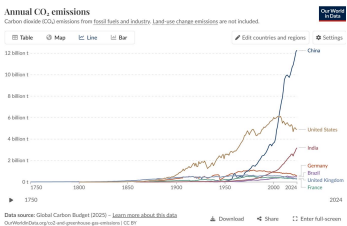
>25% of global emissions

United States

Down from ~6 Bt peak near 2005

India

~3 Bt and rising fast

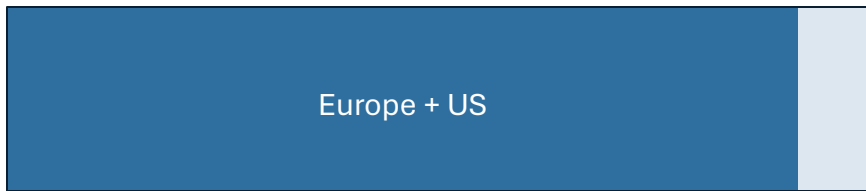


Visual reference: figure_3.jpg

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900

Europe + US: >90% of global emissions



1950 checkpoint

Europe + US still represented >85%

Comparison visual: side-by-side stacked shares; no unreported country-level split inferred.

regional shift



2024

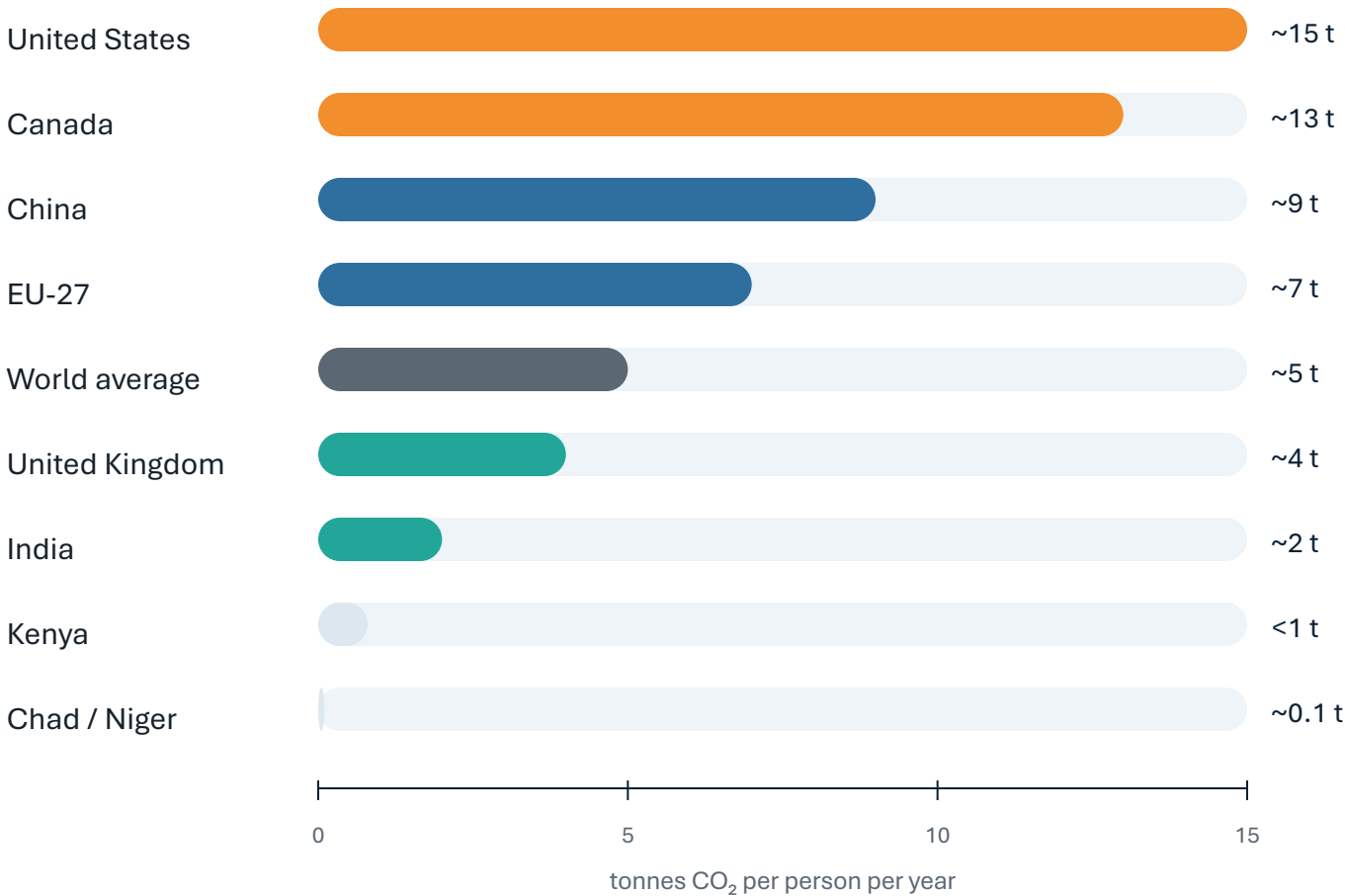
Asia: ~50%; US + Europe: < one-third



The geographic center of emissions has shifted dramatically toward Asia, while Africa and South America each contribute only 3–4%.

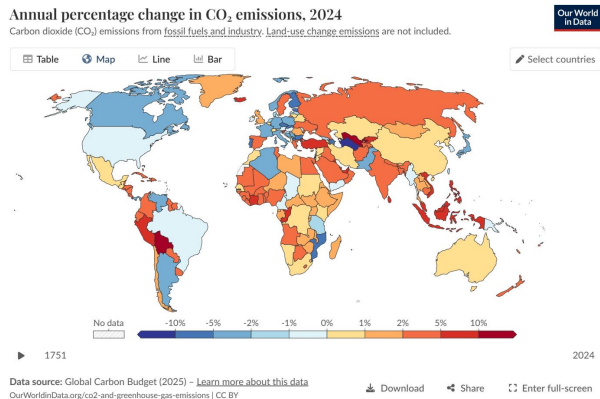
Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

Selected per capita CO₂ emissions



150× disparity

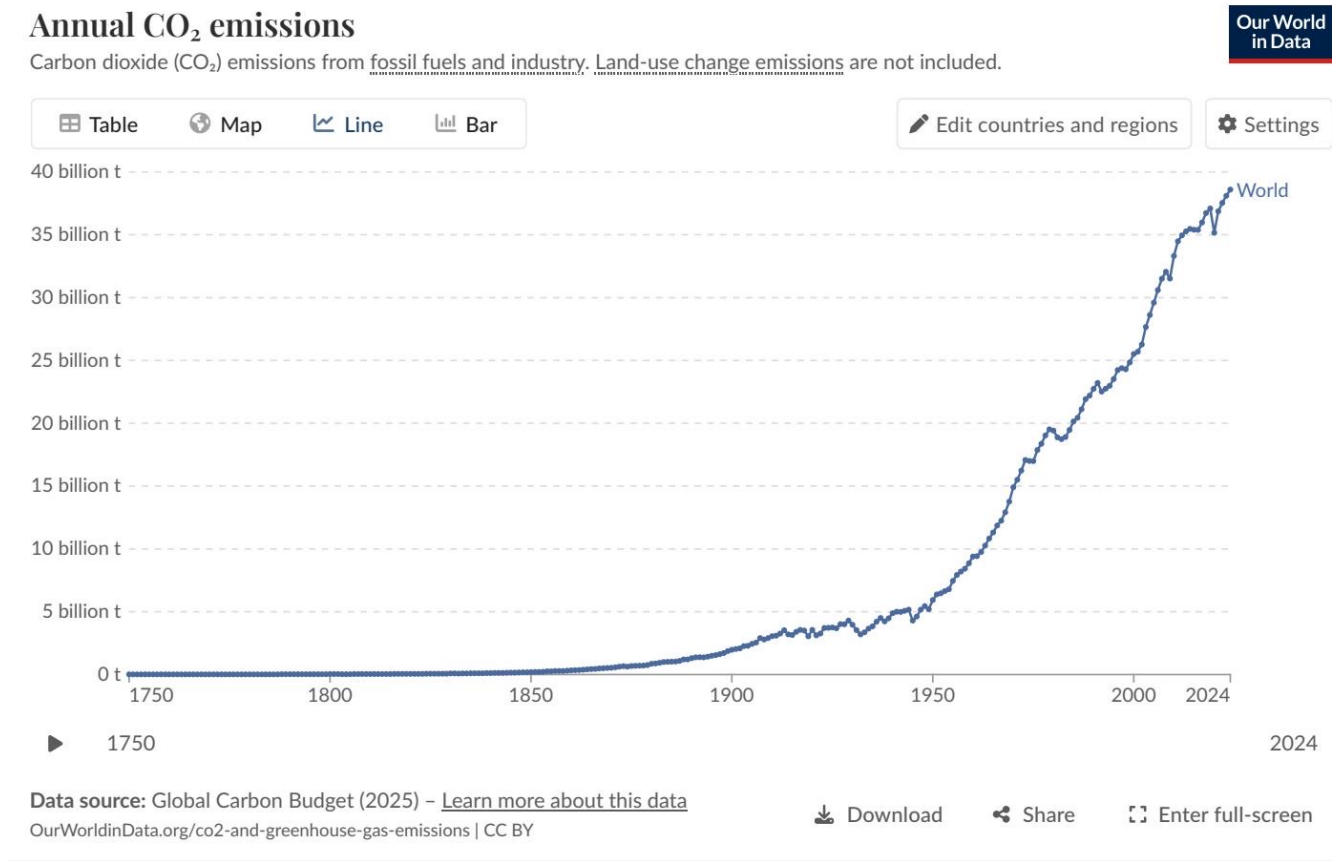
The average American or Australian emits in under two days what the average person in Chad or Niger emits in an entire year.



Reference: figure_1.jpg

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual percentage change in CO₂ emissions, 2024



Map reading

Declines

US and much of Europe: roughly -1% to -5%

Increases

Russia, parts of Southeast Asia, and several African nations

Large percentage moves

Some South American countries exceed +10%; others c

Reference: figure_4.jpg

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Same prosperity, different emissions

Countries with similar living standards can differ significantly in per capita emissions depending on their energy sources. France and the UK have much lower per capita emissions than Germany or the Netherlands, largely because of higher nuclear and renewable shares.

Policy implication

Energy mix choices shape per capita outcomes.

Energy mix comparison from source brief

Countries	Energy source pattern	Per capita CO ₂
France / UK	Higher nuclear and renewable shares	France ~4 t; UK ~4 t
Germany / Netherlands	Higher fossil fuel share	Higher than France / UK

Caution: source quantifies France and UK only; Germany/Netherlands are directional comparisons.

Key Takeaways

Emissions are still rising, but the map of responsibility is shifting

1

Global emissions exceed 35 Bt/year and have not yet peaked.

2

China accounts for >25% (~12.5 Bt), more than double the US (~5 Bt).

3

The geographic center shifted from >90% Europe+US in 1900 to Asia ~50% today.

4

Per capita inequality remains extreme: a 150× gap between high and lowest emitters.

5

Policy and energy mix choices — not just prosperity — determine per capita outcomes.